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[Frontispiece : portrait of Professor Cayley, reproduction by A. G. Dew-Smith, 1885.]

THE COLLECTED
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OF

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ADVERTISEMENT.

The present volume contains 93 papers numbered 706 to 798, published, with the exception of one series, for the most part in the years 1878 to 1883. This series is constituted by the articles which Professor Cayley wrote for the *Encyclopædia Britannica* between the years 1878 and 1888; it seemed desirable to place these together in the same volume, in spite of the departure from the chronological arrangement which governs the sequence of the papers in the volumes generally. The Syndics of the University Press desire to acknowledge their obligation to Messrs Adam and Charles Black, Publishers of the ninth Edition of the *Encyclopædia Britannica*, for their courteous consent in allowing these articles to be included in the *Collected Mathematical Papers.* Exact references to the volumes, from which the articles are extracted, will be found in the Table of Contents.

The frontispiece to the present volume is a reproduction by Mr A. G. Dew-Smith, of Trinity College, of a photograph of Professor Cayley which he made in the year 1885. The Syndics of the Press desire to acknowledge their obligation to Mr Dew-Smith.

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A. R. FORSYTH.

21 November, 1896.

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706.

ON THE DISTRIBUTION OF ELECTRICITY ON TWO SPHERICAL SURFACES.

[From the *Philosophical Magazine*, vol. v. (1878), pp. 54—60.]

IN the two memoirs "Sur la distribution de l'électricité à la surface des corps conducteurs," *Mém. de l'Inst.* 1811, Poisson considers the question of the distribution of electricity upon two spheres : viz. if the radii be a, b, and the distance of the centres be c (where $c > a + b$, the spheres being exterior to each other), and the potentials within the two spheres respectively have the constant values h and g, then—for Poisson's $f\left(\dfrac{x}{a}\right)$ writing $\phi(x)$, and for his $F\left(\dfrac{x}{b}\right)$ writing $\Phi(x)$—the question depends on the solution of the functional equations

$$a\,\phi(x) + \frac{b^2}{c-x}\,\Phi\left(\frac{b^2}{c-x}\right) = h,$$
$$\frac{a^2}{c-x}\,\phi\left(\frac{a^2}{c-x}\right) + b\,\Phi(x) = g,$$

where of course the x of either equation may be replaced by a different variable.

It is proper to consider the meaning of these equations : for a point on the axis, at the distance x from the centre of the first sphere, or say from the point A, the potential of the electricity on this spherical surface is $a\,\phi x$ or $\dfrac{a^2}{x}\,\phi\left(\dfrac{a^2}{x}\right)$, according as the point is interior or exterior; and, similarly, if x now denote the distance from the centre of the second sphere (or, say, from the point B), then the potential of the electricity on this spherical surface is $b\,\Phi x$ or $\dfrac{b^2}{x}\,\Phi\left(\dfrac{b^2}{x}\right)$, according as the point is interior or exterior; $\phi(x)$ is thus the same function of